CP-MCQA

CP-MCQA[51]

Steps:

50% of the data

Calibration Set

$S(x, Y) = 1 - [f(x)]_y$ ⇒ A classifier outputs softmax probability for each class

Calculate each sample's score

$S_1, S_2, \ldots, S_n$ (using the score of the ground-truth class)

Sort from small to large

α means the true label has at least 1−α probability of appearing in the prediction set C(x). Set to 0.1 in the paper.

$$\hat{q}_\alpha = \text{Quantile}\left(\{S_1, \ldots, S_n\}, \frac{(n+1)(n-\alpha)}{n}\right)$$

During inference

$$C(x) = y : S(x, y) \leq \hat{q}_a$$

$|C(x)|$ represents uncertainty

Example:
In Calibration Set

S_1	S_2	S_3	S_4	S_5
0.1	0.4	0.25	0.5	0.45

$\hat{q}_{0.1} \approx 0.45$

C(x) = {Cat}
|C(x)|=1 indicates low uncertainty

Class	$f(x)_y$	$S(x, y)$
Cat	0.6	0.4
Dog	0.3	0.7
Bird	0.1	0.9

[51] Conformal Prediction with Large Language Models for Multi-Choice Question Answering, ICML 2023

CP-MCQA[51]

- **Task Type:** Multiple-choice QA

- **Introduction:** Conformal Prediction for Multi-Choice Question Answering (CP-MCQA) filters out a set of predicted answers by calculating a softmax score, ensuring that the true answer lies in this set with a user-specified probability of $1-\alpha$. It quantifies the uncertainty of the prediction by the size of the prediction set.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Token-level	White-box	Post-hoc	Low	Coverage/Ranking

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